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Multimodal Concept Bottleneck Models

Dual Concept Layers Make CLIP Fully Interpretable Without Sacrificing Accuracy—or Requiring Retraining for New Classes

By Tongqing Shi, Ge Yan, Tuomas Oikarinen & Tsui-Wei Weng (UC San Diego)

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Interpretable image classifiers no longer need to sacrifice accuracy or be retrained for every new class. MM-CBM introduces dual Concept Bottleneck Layers for both the image and text branches of CLIP, forcing predictions to pass through a human-readable concept space while matching black-box CLIP accuracy within 5%—and enabling zero-shot transfer to unseen classes with no additional training.

AT A GLANCE

Problem: Existing Concept Bottleneck Models are limited to fixed label sets and suffer from concept leakage, where predictive signals outside the intended concept space are inadvertently exploited.

Why it matters: Interpretable multimodal models are critical for high-stakes domains (medical imaging, autonomous driving) where users need to understand and intervene on model reasoning.

Method: Dual Concept Bottleneck Layers (CBLs) align both image and text CLIP embeddings into a shared interpretable concept space; classification is a pure concept-space dot product with no linear head, eliminating leakage by construction.

Result: Up to 51.26% accuracy improvement over prior CBMs; within 5% of black-box CLIP; zero-shot transfer to unseen classes.

Evidence: CUB-200-2011, ImageNet, Oxford Pets, Stanford Cars; vs. Label-free CBM, Post-hoc CBM, LaBo.

The Big Picture

Deep neural networks are powerful but opaque. Concept Bottleneck Models (CBMs) are a popular interpretability approach: they insert a bottleneck layer whose activations correspond to named concepts (e.g. “has wings,” “metallic surface,” “striped”), making the model’s reasoning transparent. A user can inspect which concepts drove a prediction and directly intervene by modifying concept

scores.

Standard CBMs have two structural limitations. First, they are trained for a fixed label space. If a new class appears, the model must be retrained. Second, the final classification head (a linear layer over concept scores) can exploit non-concept information encoded in concept activations—a phenomenon called *concept leakage*. The model appears interpretable but silently uses signals invisible to the user.

MM-CBM addresses both problems by extending CBMs into CLIP’s shared vision-language embedding space. Two Concept Bottleneck Layers—one for image embeddings, one for text embeddings—project both modalities into a unified concept space. Classification is then a dot product between the image concept vector and the text concept vector for each class description. No separate classification head is needed; leakage is impossible because the concept space fully mediates inference. New classes require only their text description—no retraining.

Why Existing Approaches Fall Short

Label-free CBM: Uses CLIP to identify concept-image alignments without manual concept labels, but still trains a linear head, preserving leakage.

Post-hoc CBM: Applies concept bottleneck post-training by projecting frozen representations onto concepts. Accuracy drops significantly on fine-grained datasets.

LaBo: Uses language models to generate concepts and assembles a bottleneck via logistic regression. Fixed to training classes; concept leakage present via the regression head.

None of these methods operate purely in the concept space, and all require retraining or re-optimisation for new classes.

Core Mechanism

Let K concept descriptions $\{c_1, \dots, c_K\}$ be given (e.g., from a domain expert or generated by an LLM). Compute concept text embeddings $e_k = f_T(c_k)$ using CLIP’s text encoder f_T .

For an image x_v with CLIP embedding $z_v = f_V(x_v)$, the image concept vector is:

$$[c_v]_k = \frac{\langle W_v z_v, e_k \rangle}{\|W_v z_v\| \cdot \|e_k\|}, \quad k = 1, \dots, K,$$

where $W_v \in \mathbb{R}^{d \times d}$ is the learnable image CBL projection. The text concept vector for class l_j is:

$$[c_t(l_j)]_k = \frac{\langle W_t z_t(l_j), e_k \rangle}{\|W_t z_t(l_j)\| \cdot \|e_k\|}.$$

Classification: $\hat{y} = \arg \max_j \langle c_v, c_t(l_j) \rangle$. No linear head; no leakage.

Technical Formulation

Training objective. Let $\mathcal{D}_{\text{aux}}$ be an auxiliary dataset of image-text pairs. The loss combines three terms:

$$\mathcal{L} = \mathcal{L}_{\text{align}} + \lambda_{\text{task}} \mathcal{L}_{\text{task}} + \lambda_{\text{sparse}} \|c_v\|_1.$$

$\mathcal{L}_{\text{align}}$ is a CLIP-style contrastive loss on (c_v, c_t) pairs:

$$\mathcal{L}_{\text{align}} = -\frac{1}{B} \sum_{b=1}^B \log \frac{e^{\langle c_v^b, c_t^b \rangle / \tau}}{\sum_{b'} e^{\langle c_v^b, c_t^{b'} \rangle / \tau}}.$$

$\mathcal{L}_{\text{task}}$ is cross-entropy on training classes, using c_v as logits (after matching against class concept vectors).

$\|c_v\|_1$ is an L_1 sparsity penalty ensuring that each prediction uses only a small number of active concepts.

Zero-shot transfer. For a new class l_{new} , compute $c_t(l_{\text{new}})$ using the text encoder and CBL. No retraining required. This follows from the universal concept space shared by image and text CBLs.

User intervention. To intervene on concept k : set $[c_v]_k \leftarrow v$ for any desired value $v \in [-1, 1]$. The corrected concept vector propagates directly to the prediction via the dot product—fully transparent.

Learning or Inference Procedure

Training:

1. Freeze CLIP encoders f_V, f_T .
2. Initialise W_v, W_t as identity.
3. Train on $\mathcal{D}_{\text{aux}}$ minimising $\mathcal{L}$ with Adam optimiser (5×10^{-4} lr, 50 epochs).

Inference: two forward passes (f_V and W_v), one dot product per class. Complexity $O(K \cdot d)$ per image—faster than CLIP zero-shot.

What the Guarantee Says

By construction, all predictive signal in MM-CBM flows through the concept vectors c_v and c_t . Since the final classifier is a dot product in the concept space (no additional parameters), the model cannot encode information outside the K -dimensional concept subspace. This is a **zero-leakage** guarantee: any prediction not expressible via concept scores is structurally impossible.

For zero-shot transfer, the guarantee is that any class expressible via concept descriptions $\{c_k\}$ is immediately classifiable. Accuracy on unseen classes depends on how well the concept vocabulary spans the relevant visual dimensions—a design choice, not a training variable.

Experimental Findings

- **CUB-200-2011:** MM-CBM achieves 78.3% vs. 53.1% for Label-free CBM (a 47.5% relative gain) while staying within 4% of black-box CLIP.
- **ImageNet:** Within 5% of CLIP zero-shot; best interpretable method.
- **Oxford Pets / Stanford Cars:** Consistent improvements over all CBM baselines (average 51.26% accuracy improvement).
- **Intervention:** Correcting one concept score improves accuracy by 12% on targeted error cases.
- **Zero-shot:** Correctly classifies 8 unseen bird species added at test time, with no accuracy degradation on original classes.

Ablations and Interpretation

Remove image CBL ($W_v = I$): Accuracy drops by ≈ 8 percentage points; concept alignment is worse.

Remove text CBL ($W_t = I$): Similar drop; text concept vectors do not align with visual concept space without projection.

Remove sparsity loss: Accuracy +0.3% but average concept activations per prediction increase $8\times$, making interpretations unintelligible.

Concept set size K : Performance plateaus at $K = 100$; smaller sets ($K < 30$) significantly hurt accuracy on fine-grained tasks.

Reference

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